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Quantification and the effects of secondary colostrum components on neonatal calf physiology and metabolism: A scoping review

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ABSTRACT

Bovine colostrum is produced in the weeks leading up to and secreted by the dam after parturition, serving as a complete source of nutrition for neonatal calves. Research has long emphasized the importance of timely colostrum feeding, focusing primarily on IgG transfer and the assessment of passive immunity within 24 to 48 h of intake. However, emerging evidence indicates that "secondary" components—cytokines, oligosaccharides, fatty acids, enzymes, and other nutrients—also play critical roles in metabolism and physiological development. A further examination and understanding of their dynamics could refine colostrum management strategies and support targeted approaches, such as extended colostrum feeding or inclusion of transition milk, to improve neonatal outcomes. Despite their importance, data on these components across successive milkings remain limited, representing a key research gap. In this scoping review, 15 sources published between January 2000 and April 2025 were identified through database searches, with exclusions applied to papers not meeting the inclusion criteria. Collectively, these studies reported measures of secondary colostrum components, describing their kinetics from colostrum to transition milk and mature milk, each serving distinct biological functions in calf development. Unlike mature milk, colostrum and transition milk are uniquely rich in IgG, IgA, and IgM, as well as cytokines and oligosaccharides that promote gut health and protect against pathogens. These metabolites decrease from colostrum to the seventh milking, as shown in the literature. Growth factors like IGF-I and TGF-β are present at significantly higher concentrations in colostrum and decrease exponentially over time; studies have shown that these growth factors enhance GI tract epithelial growth and nutrient absorption. Hormones such as prolactin and cortisol also decrease exponentially after the first milking, and studies have shown that they are related to metabolic regulation. The results of this scoping review suggest that although considerable research has examined IgG, IgA, and IgM levels, as well as cytokines and oligosaccharides and other colostrum bioactive compounds from first to mature milking, few studies have comprehensively quantified other components. Also, most studies have investigated a single component or related components and none have investigated holistically the secondary components in colostrum, transition milk, and mature milk. Also, no study has investigated the effects of management and other factors in the transition milk of dairy cows, underscoring the need for further investigation.

Key words: colostrogenesis, passive immunity, oligosaccharides, transition milk

INTRODUCTION

Dairy calves are born agammaglobulinemic and require the passive transfer of immunity via the ingestion of colostrum (Weaver et al., 2000). A ruminant develops a cotyledonary placenta during gestation, which does not permit passive transfer of immune factors in utero to the fetus(es). Colostrum contains several factors essential for the survival of the calf, including macronutrients, micronutrients, and secondary components, the latter of which will be the focus of this literature review. The current landscape of colostrum understanding on dairy farms in the United States was reported by the National Animal Health Monitoring System (USDA, 2021a). They summarized work emphasizing importance on quality, quantity, timeliness of feeding, and quantifying passive immunity status of dairy calves after feeding colostrum. Most importantly, a benchmark for colostral and serum immunoglobulin G (IgG) was established, which has contributed to decreased morbidity and mortality on farms (Lombard et al., 2020). Furthermore, a collaborative study determined morbidity and mortality rates in 2014 were 33.9% and 5.0% (USDA, 2021b) compared with higher incidence rates reported in 1994 (USDA, 1994) and 2007 (USDA, 2007), when feeding of colostrum to individual calves and ranges for

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colostrum feeding were both reported to be lower. These documents identify improvements to the overall care and management of dairy calves and implemented several steps to ensure successful passive transfer of immunity on dairy farms in the United States. Tasked with caring for these animals and achieving high standards of care, some work has been done to understand how to better encourage these efforts. Wilson et al. (2023) described the role of benchmarking as a potential tool to improve or maintain care of animals specific to colostrum feeding practices. A big area of concern is the value ascribed to male versus female calves and, in the case where a calf is not intended for replacement, the reduction in care that might occur for those calves deemed to have less value (Creutzinger et al., 2022).

Over the past decade, studies have addressed dry cow management and determining the best feeding practices for improving colostrum quality and quantity produced by the dam (i.e., Mann et al., 2016; Westhoff et al., 2024b), identifying the effects of harvesting and preservation processing on colostrum quality (i.e., Stewart et al., 2005; Abdelsattar et al., 2022; Denholm, 2022), looking at the delivery and absorption of colostrum components by the calf (Conneely et al., 2014; Desjardins-Morrissette et al., 2018; Lopez et al., 2020), and complement feeding colostrum after the first 24 to 48 h of life (i.e., Van Soest et al., 2020; Zwierzchowski et al., 2020; Bahadori-Moghaddam et al., 2023). Today, components of colostrum are garnering increased attention, with several reviews highlighting their importance when fed to offspring in their respective species, as well as the effects of these components when harvested and used for human consumption (i.e., Fischer-Tlustos et al., 2021; Poonia and Shiva, 2022; Silva et al., 2024).

The eighth edition of Nutrient Requirements of Dairy Cattle is a critical text summarizing key data related to the feeding of dairy cows, heifers, and calves (NASEM, 2021). Notably, the text highlights several important factors related to colostrum and the growing interest in understanding how secondary compounds are produced by the dam and their effects on neonatal calf development when ingested via colostrum. In this scoping review, our overall objective was to systematically map, quantify, and describe the available evidence, identify gaps, and clarify limitations in the literature around "secondary" components of colostrum, such as cytokines, oligosaccharides, fatty acids, enzymes, and other nutrients, and how they interact and play critical roles in metabolism and physiological development of the neonate dairy calf. Specifically, our first objective was to highlight the process of colostrogenesis and the management factors that influence colostrum and transition milk composition. Second, we aimed to perform a systematic collection of quantitative values for secondary colostrum components from journal articles published between 2000 and 2025 and summarize the data to highlight their natural composition and kinetics. When available, we compared concentrations across subsequent milkings. Finally, we discuss the effects of these components on neonatal calf development based on the available literature and highlight gaps in the literature.

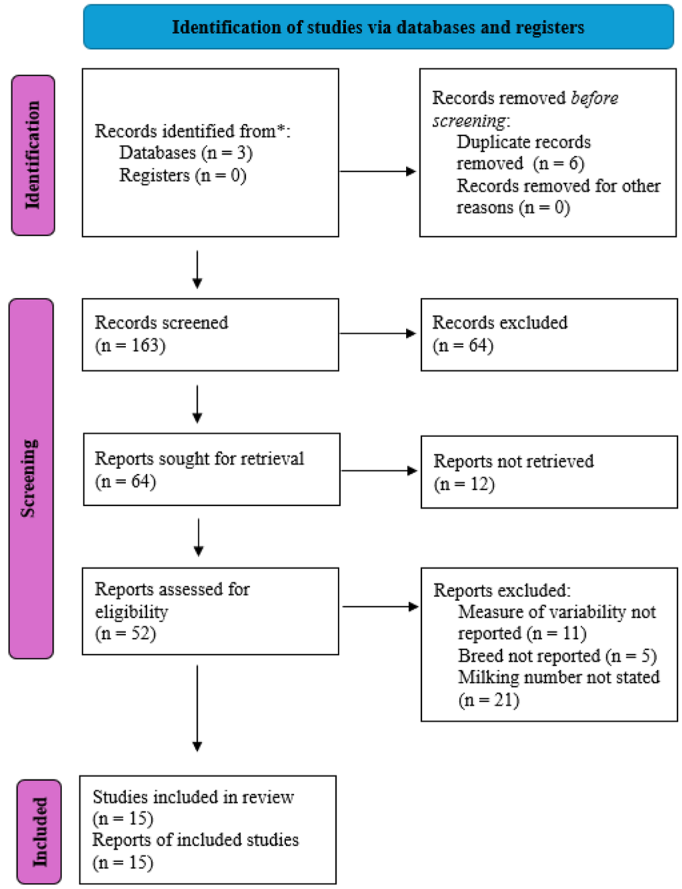

Figure 1. Study selection process. A literature search was conducted using the terms to identify relevant studies examining secondary components of colostrum. The search included publications from 2000 to 2025. The term "n" indicates the number of studies included or excluded at each stage of the selection process.

MATERIALS AND METHODS

A scoping review of the literature was conducted to identify secondary colostrum component quantities in colostrum, transition milk, and mature milk. To conduct this review, we established a protocol (Figure 1) following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Page et al., 2021). A search of electronic databases including Google Scholar (https://scholar.google.com/), PubMed (https:

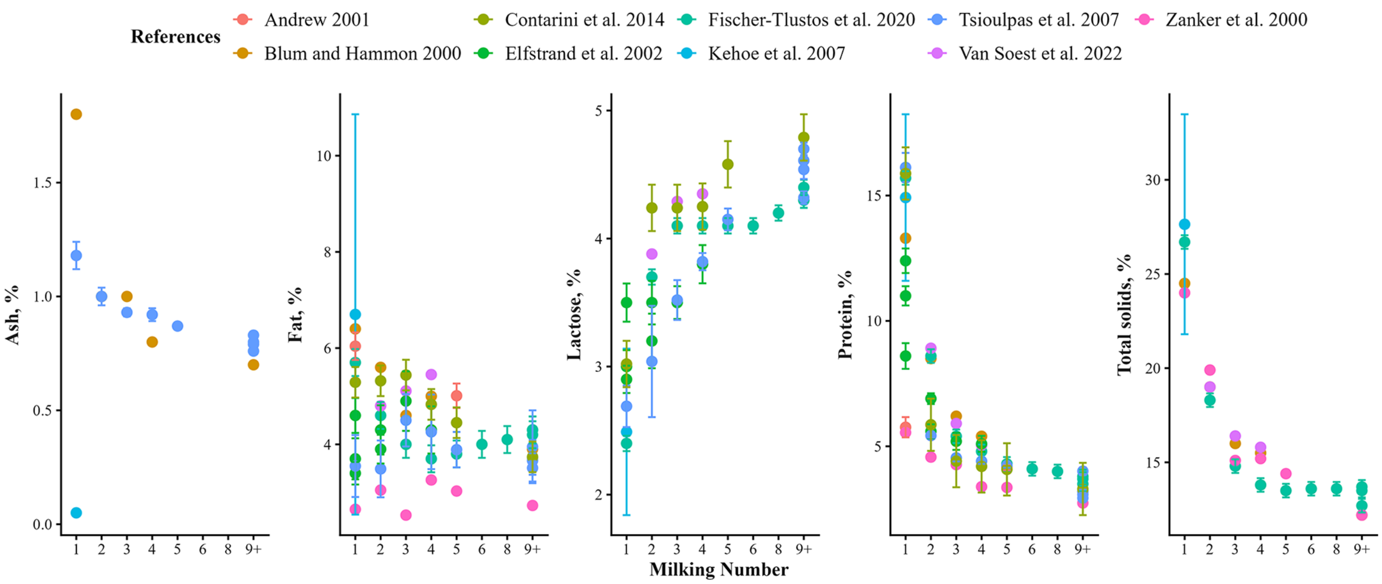

Figure 2. Gross composition of bovine colostrum, transition milk, and mature milk over time from a literature search from 2000 to 2025 (n = 9 studies included). Components from milk extracted and presented here as percentages: ash, fat, lactose, protein, and TS. Error bars represent SE. References: Blum and Hammon (2000); Zanker et al. (2000); Andrew (2001); Elfstrand et al. (2002); Kehoe et al. (2007); Tsioulpas et al. (2007); Contarini et al. (2014); Fischer-Tlustos et al. (2020); Van Soest et al. (2022).

//pubmed.ncbi.nlm.nih.gov/), Science Direct (https://www.sciencedirect.com/) was conducted using strings of search terms related to colostrum components using the terms "bovine colostrum" in combination with "vitamins and minerals," "enzymes," "fatty acids," "oligosaccharides," "cytokines," "amino acids and proteins," "hormones," and "growth factors." The articles included in this meta-analysis were required to be published in English, with data collected between January 2000 and April 2025. This 25-year window was selected to capture developments over the past quarter century, a period marked by significant advancements in dairy production, including improvements in analytical technologies, breeding strategies, and calf management practices.

Identified articles (n = 15) were imported to Mendeley (version 2.132.2, Elsevier, Mendeley Ltd.), and duplicates were removed. One reviewer was tasked with screening the article and determining based on title and abstract if the experiment collected colostrum from cows with no major nutritional or health interventions before parturition and reported the concentration of the secondary colostrum component. If the article met these criteria, then the study underwent a full-text review to determine if the researchers quantified secondary colostrum components by milking number, identified the breed of the cow, reported observation count for component analysis, and reported a measure of variability (SD or SE).

Data were extracted and recorded into MS Excel manually by one researcher from tables and figures, and by WebPlotDigitizer (V.5.2; Rohatgi, 2024) to estimate numbers from figures where none were provided. The Excel sheet was subsequently uploaded in R (V. 4.4.1; https://www.r-project.org) for data visualization.

RESULTS AND DISCUSSION

Colostrogenesis and Nutritional and Management Effects on Bovine Colostrum Composition

Colostrogenesis is the active transfer of IgG and bioactive components into mammary secretions the days or weeks before parturition (Baumrucker et al., 2010; Baumrucker and Bruckmaier, 2014; Baumrucker et al., 2021). During this process, a mass transfer of IgG into colostrum occurs and after the second milking, there is a reduction in how many IgG are present. The transition from colostrum to mature milk occurs gradually over several milkings after parturition and is quantified by the gross composition differences of colostrum, transition milk, and mature milk (Figure 2).

In this scoping review, colostrum refers to the first milking collected within 24 h after birth. Transition milk includes milkings from d 2 to 3 postpartum (milkings 2 to 6), and milk produced thereafter (milking 7 and beyond) is generally considered mature milk. Colostrum is different from mature milk because it contains more protein, sodium, and chloride, and less potassium and lactose compared with mature milk (Baumrucker et al., 2021). The SCC is much higher due to leaky tight junctions of the mammary gland permitting a mass transfer of cells at

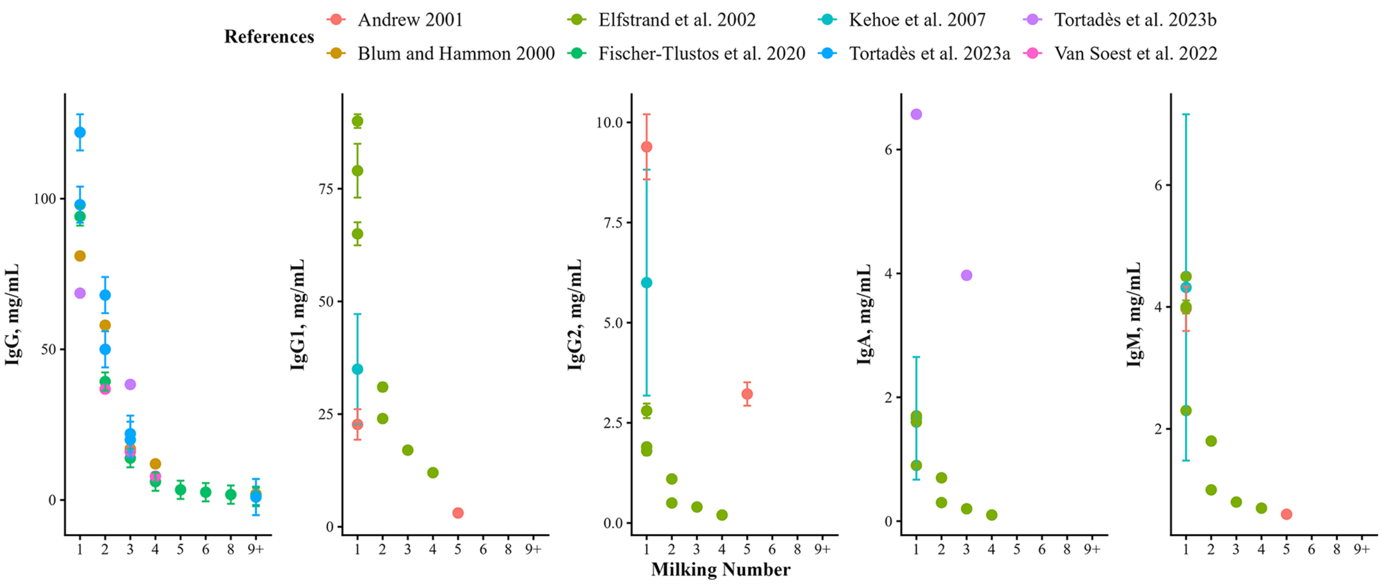

Figure 3. Immunoglobulin content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 8 studies included). Components from milk extracted and presented here (in mg/mL) IgG, IgG1, IgG2, IgA, and IgM. Error bars represent SE. References: Blum and Hammon (2000); Andrew (2001); Elfstrand et al. (2002); Kehoe et al. (2007); Fischer-Tlustos et al. (2020); Van Soest et al. (2022); Tortadès et al. (2023a,b).

this stage before parturition (Nguyen and Neville, 1998; Baumrucker et al., 2021). Several pathways are responsible for the transfer of components to colostrum, more recently the bovine Fc receptor (FcRn) garnered attention because it is mainly attributed to IgG accumulation in colostrum (Baumrucker et al., 2021). Today, there are specific markers characterizing the differences between colostrum and mature milk based on proteomics, fatty acids, and more (Fahey et al., 2020; Wilms et al., 2022).

The amount of energy provided in the prepartum diet may influence colostrum composition (Mann et al., 2016; Richards et al., 2020; Vasquez et al., 2021). Mann et al. (2016) reported that higher-energy diets in the dry period were associated with lower colostral IgG concentrations, higher colostral insulin, and increased de novo fatty acids. In contrast, Richards et al. (2020) and Vasquez et al. (2021) observed no effect of prepartum dietary energy on colostrum yield or IgG. However, in both studies, the dietary treatments were adjusted for proportional DMI. As a result, cows consuming higher-energy diets reduced their DMI, leading to similar total energy intake across treatments and likely masking any potential effects on colostrum composition. Furthermore, MP supply changed volume of colostrum for cows in parity 2 and increased fat yield (Westhoff et al., 2024a). Several recent trials have shown a positive effect of choline supplementation on colostrum yield or quality (Zenobi et al., 2018; Swartz et al., 2022; Holdorf et al., 2023). In addition, rumen-protected Arginine had an effect on colostrum volume (Souza Simões et al., 2024).

In a comparative study by Kessler et al. (2020), both breed and parity significantly influenced colostrum IgG concentration. Notably, this study highlighted substantial individual variation in colostrum production, suggesting that factors beyond breed and parity play a critical role in determining IgG levels. In another study, parity differences influence the composition of bioactive components in milk (Fischer-Tlustos et al., 2024). This study examined a small cohort of cows (n = 10 multiparous and n = 10 primiparous) to assess differences in insulin, IGF-I, and lactoferrin concentrations across successive milkings during the transition from bovine colostrum to whole milk. The IGF-I and insulin levels declined rapidly, while lactoferrin decreased more gradually. Parity affected both insulin and lactoferrin: Primiparous cows had higher insulin concentrations and yields, whereas multiparous cows had higher lactoferrin yields only. Additionally, multiparous cows showed associations between bioactive components and mammary adaptations resulting from repeated lactations. Parity also affects mineral abundance in bovine colostrum (Kume and Tanabe, 1993). Colostral calcium, phosphorus, and magnesium concentrations were lower in multiparous cows compared with primiparous cows at parturition.

Oxytocin is known to influence colostrum quantity and, in some cases, its quality (Sutter et al., 2019; Mann et al., 2024). Sutter et al. (2019) did not observe an increase in colostrum volume when comparing cows separated from their calf, cows in the presence of their calf, and cows administered 20 IU of oxytocin. Howev-

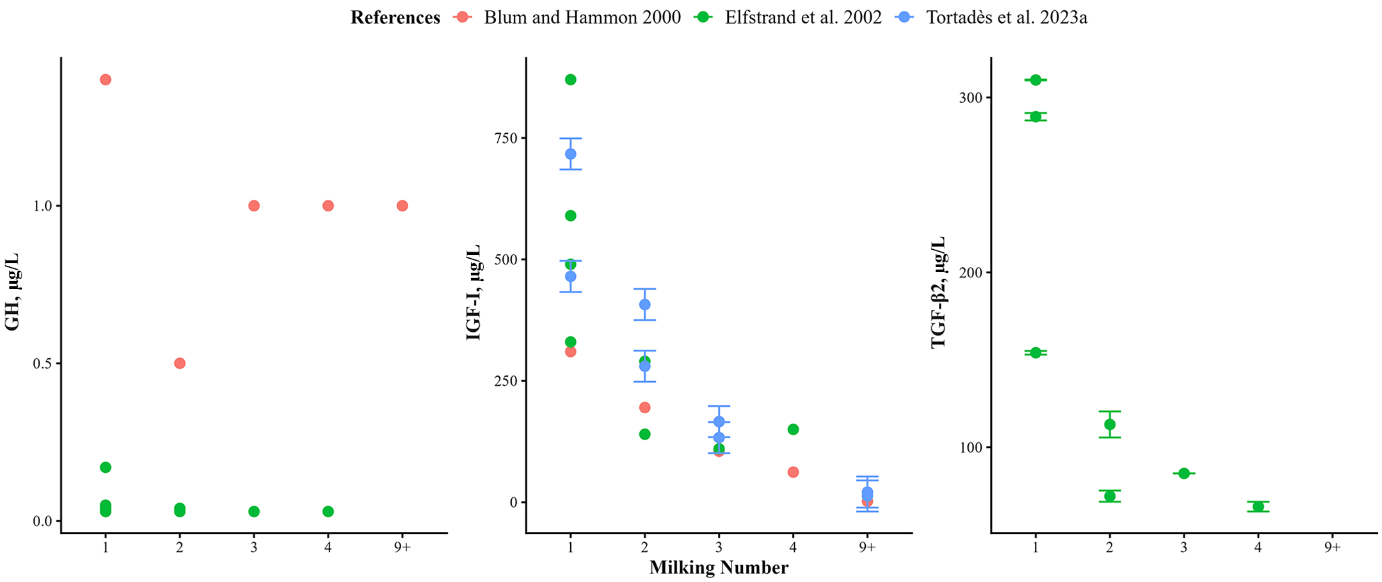

Figure 4. Growth factor content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 3 studies included). Components from milk extracted and presented here (in µg/L) include growth hormone (GH), IFG-I, and TGF-β2. Error bars represent SE. References: Blum and Hammon (2000); Elfstrand et al. (2002); Tortadès et al. (2023a).

er, they did observe higher IgG concentrations in cows given oxytocin and in cows with their calf present. In contrast, Mann et al. (2024) reported that administering 40 IU of oxytocin (compared with 0 or 20 IU) increased colostrum volume in primiparous cows without altering colostrum composition. This effect was not due to a disruption of the blood-milk barrier but is likely influenced by management factors and stress, which may introduce confounding effects. Additionally, the time after the first milking affected the IgG in colostrum. Mineral status and form of minerals fed to the cow directly affects the composition of the colostrum (Van Emon et al., 2020). Typically, the minerals are transferred in utero; however, as summarized by Van Emon et al. (2020), there are correlations between the abundance of minerals in colostrum with the immunoglobulin status to which higher amounts of selected minerals show a greater amount of IgG. Organic trace minerals have been associated in some cases with increased IgG, Ca, and Mg in colostrum compared with inorganic forms. Selenium supplementation, particularly in organic form, consistently increased colostral Se concentrations, although effects on IgG are inconsistent. Copper and zinc supplementation have shown variable effects on colostrum composition, often dependent on baseline mineral status, with zinc sometimes associated with higher IgG concentrations. Some minerals, such as Mn, Cu, and Zn, may increase colostrum volume without altering composition.

Given the inherent variability in colostrum composition and the potential influence of maternal interventions during the transition period, it is essential to consider the broader factors that contribute to these changes. A comprehensive understanding of these influences provides valuable context for interpreting colostrum variability. Building on this foundation, the following sections review findings from a range of studies that have investigated the composition of colostrum and associated components.

Quantification of Immunoglobulins and Non-IgG Components in Colostrum and Effects of Secondary Components on the Physiology of the Neonatal Calf

Before exploring the absorption and effects of secondary components on neonatal calf development, it is important to consider the role of fetal programming in shaping neonatal outcomes and ability to utilize these components. Fetal programming—affected by maternal factors such as environment, stress, and health status before parturition—can significantly influence calf physiology, including the efficiency of colostral component absorption (Osorio, 2020). Heat stress of the dam reduces fetal growth and supply of AA in utero (Tao and Dahl, 2013; Wu et al., 2019). When a calf is delivered preterm, intestinal morphology is compromised, showing a lack of maturity and a reduction or absence of receptors responsible for gut maturation (growth factors), including receptors for IGF-I and IGF-II, along with disruptions in hormone secretion (Bittrich et al., 2004). Moreover, blood mineral profile of calves with stunted growth in utero, exhibited compromised mineral status (Chernitskiy et al., 2022).

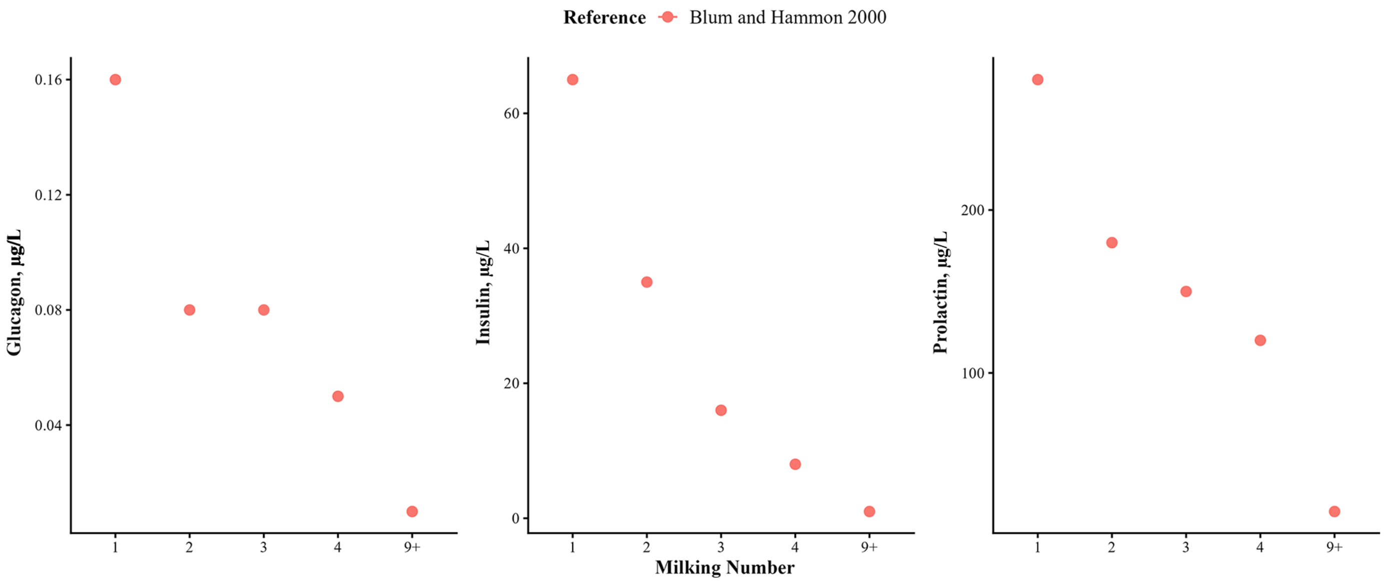

Figure 5. Hormone content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 1 study included; Blum and Hammon, 2000). Components from milk extracted and presented here (in µg/L) include glucagon, insulin, and prolactin.

The maturation and development of the neonatal calf gastrointestinal tract (**GIT**) morphology are characterized by an increase in ileal and jejunal crypt depth, as well as a progressive rise in duodenal villus height over time following birth (Guilloteau et al., 2009; Meale et al., 2017). Changes in gut morphology, including increasing abomasal and mucosal weight, affects the absorption of components absorbed in these areas. The environment of the neonatal calf is categorized as a functional monogastric and the abomasum and intestine are the prime digestion sites (Davis and Drackley, 1998). Overall enzyme activity (pepsin and amylase) is decreased in the neonatal calf gut, and HCl levels are low to promote the absorption of factors specific to the neonatal period, which dynamically change over time (Guilloteau et al., 1984, 2009).

The concept of gut closure is the underlying mechanism for the absorption of Ig where intestinal permeability ceases after 12 to 24 h (Stott et al., 1979). Stott et al. (1979) identified gut closure by measuring the Ig concentration in neonatal calf serum in 4-h increments after feeding different volumes of colostrum and identifying the different absorption rates of IgG, IgA, and IgM. Although this study measured serum Ig only up to 24 h, it solidified gut closure and lends itself to the major component of timely feeding of colostrum. More work on gut closure continues to precipitate from this work and investigation into the other components affected by gut closure are mentioned later in this review. Gut closure and delaying colostrum feeding affects passive transfer and intestinal bacteria colonization because of the reduction of pinocytotic absorption of molecules decreasing after 7 d of life (Fischer et al., 2018; Ma et al., 2019; Song et al., 2022). Furthermore, a major macronutrient present in colostrum is the protein casein, which affects absorption of nutrients (primarily fat associated secondary components) because its interaction with chymosin reduces passage rate (due to curd formation) of colostrum through the calf's GIT (Davenport et al., 2000).

Immunoglobulins. Immunoglobulin concentrations are significantly higher in colostrum compared with mature milk (Figure 3): IgG ranges from 68.64 to 122.00 mg/mL in colostrum compared with <2 mg/mL in mature milk; IgA from 0.9 to 6.57 mg/mL whereas it decreases over time to <1 mg/mL by the fourth milking; and IgM decreases from 2.30 to 4.50 mg/mL to <1 mg/mL by the fifth milking (Blum and Hammon, 2000; Andrew, 2001; Elfstrand et al., 2002; Kehoe et al., 2007; Fischer-Tlustos et al., 2020; Van Soest et al., 2022; Tortadès et al., 2023a,b). Overall, IgG constitutes 73% to 86% of all Ig provided by colostrum, where IgG1 is present at a much higher concentration compared with IgG2 at 22.7 to 90 mg/mL versus 1.8 to 9.39 mg/mL, respectively.

Immunoglobulin G provides systemic and mucosal immunity by neutralizing viruses and toxins, as well as clumping bacteria to make them more susceptible to phagocytosis (agglutination and opsonization; Barrington and Parish, 2001; Vlasova and Saif, 2021). The presence of IgA GIT provides local immunity but at a much lower rate compared with IgG. Similarly to IgG, IgM has a role in bacterial agglutination and opsonization of bacteria, as well activating the complement system, which is not fully developed at birth (Chase et

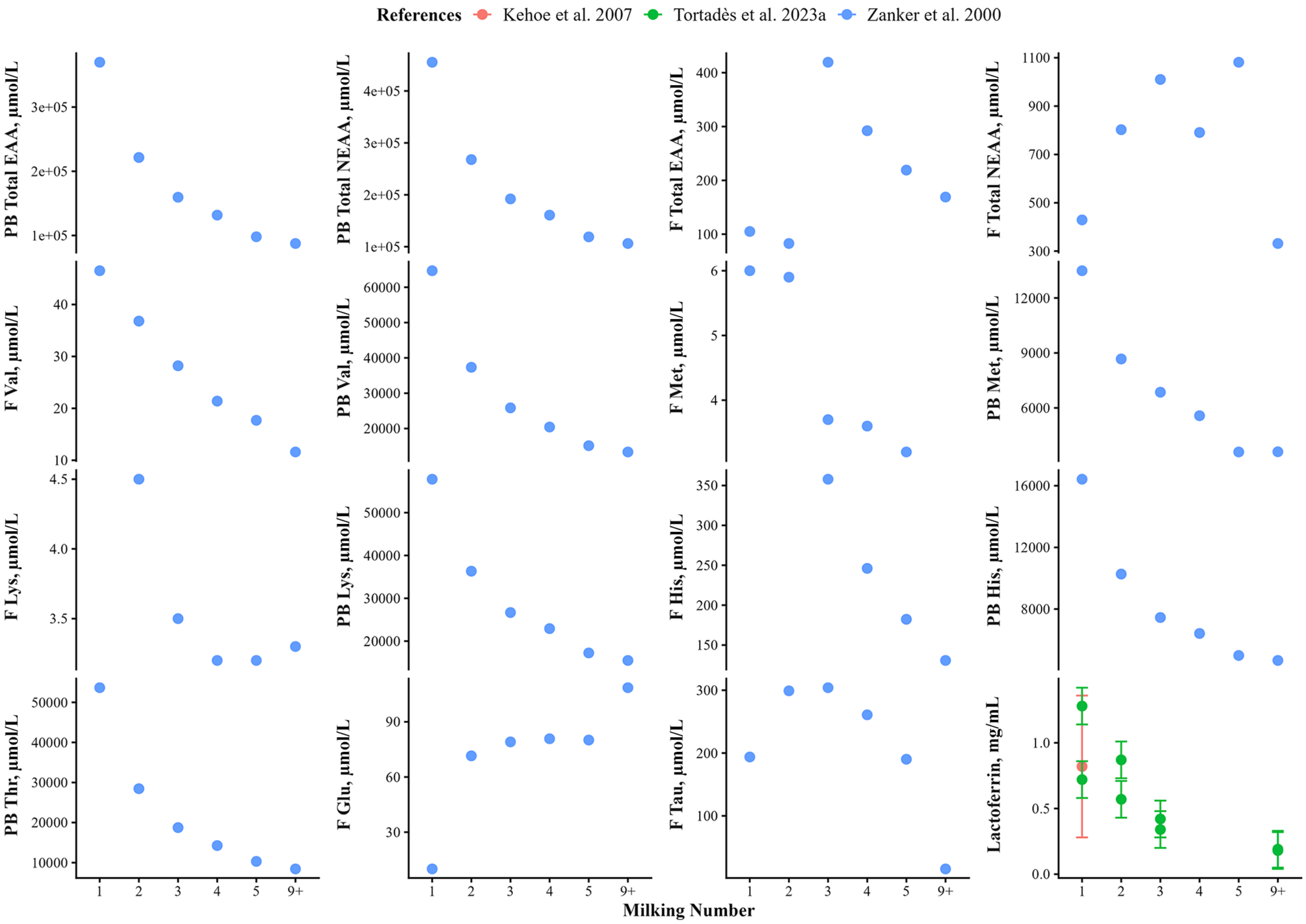

Figure 6. Selected AA and protein content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 3 studies included; Zander et al., 2000; Kehoe et al., 2007; Tortadès et al. 2023a). Components from milk extracted and presented here (in µmol/L unless otherwise noted) include PB total EAA, PB total NEAA, F total EAA, F total NEAA, F Val, PB Val, F Met, PB Met, F Lys, PB Lys, F His, PB His, PB Thr, F Glu, F Tau, and lactoferrin (mg/mL). PB = protein bound; F = free. Error bars represent SE.

al., 2008). The importance of Ig is well understood; it constitutes a large part of colostrum, playing a major role in transfer of immunity.

Growth Factors. Growth factors are present in both colostrum and transition milk, but concentrations are markedly higher in colostrum (Figure 4). Insulin-like growth factor-I concentrations range from 330 to 870 µg/L in colostrum, decreasing to 150 µg/L in transition milk. Transforming growth factor-β2 levels range from 154 to 310 µg/L in colostrum and decrease to 66 µg/L in transition milk. Growth hormone ranges between 0.03 and 0.17 µg/L in colostrum and eventually remains constant at 0.03 to 0.04 µg/L across milkings 2 to 4 (Blum and Hammon, 2000; Elfstrand et al., 2002). Importantly, the growth hormone values presented in Figure 4 highlight the limitations of hormone detection in colostrum because the very low measured concentrations likely reflect the sensitivity constraints of standard assays (Elfstrand et al., 2002).

Over time, fetal-type enterocytes are replaced by adult enterocytes, strengthening GIT integrity and reducing the passive transfer of large molecules. Insulin-like growth factor receptors are present in the jejunum and ileum (Baumrucker et al., 1994), and although IGF-I is present in colostrum, systemic effects are limited—plasma IGF-I levels do not increase beyond 24 h postbirth. The IGF contribute to the development of intestinal tissues, although they are not solely responsible for this process (Roffler et al., 2003). Their role may include reducing the activity of digestive enzymes during the neonatal period (Blum, 2006), although their function is further complicated by interactions with other hormones, IGF binding proteins, and lactoferrin. Endogenous glucagon-like peptide 2 (GLP-2), for ex-

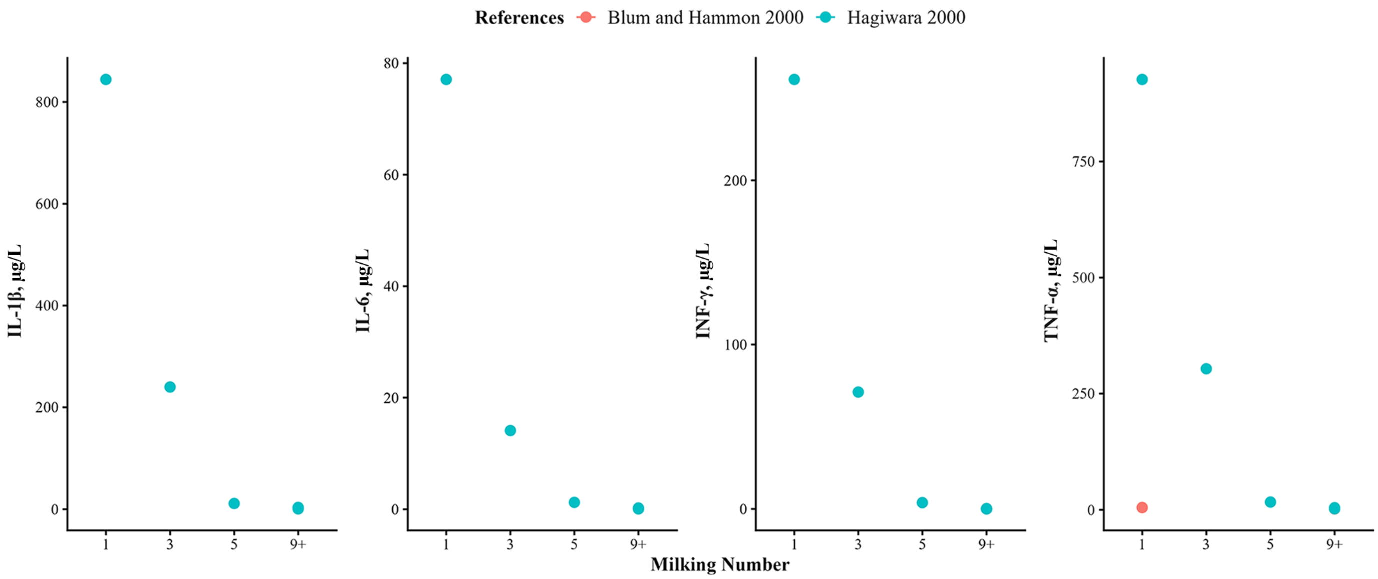

Figure 7. Cytokine content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 2 studies included; Blum and Hammon, 2000; Hagiwara et al., 2000). Components from milk extracted and presented here (in μg/mL) include IL-1β, IL-6, INF-γ, and TNF-α.

ample, is stimulated by nutrient intake and contributes to gut maturation (Burrin et al., 2003). However, the effects of exogenous hormone supplementation are mixed. In calves, growth hormone injections alongside colostrum feeding have been associated with reduced GIT development (Bühler et al., 1998), whereas in rats, similar treatment enhanced villus development in the small intestine (Gómez de Segura et al., 1996).

Natural colostral IGF-I and IGF-II stimulate epithelial cell proliferation (Baumrucker et al., 1994), although administration of recombinant IGF-I has not shown the same benefit (Bühler et al., 1998). Epidermal growth factors (**EGF**) are also known to support intestinal growth (Klagsbrun, 1980), but their concentration in bovine colostrum is not well documented compared with species with higher EGF levels (Pakkanen and Aalto, 1997; Baumrucker and Macrina, 2016). Although GLP-2 ingestion may influence epithelial proliferation, its direct effect is still uncertain (Inabu et al., 2021). Transforming growth factors (**TGF**) have been shown to enhance mucosal immunity and promote tissue growth (Gauthier et al., 2006). Taken together, growth factors in colostrum appear to support neonatal development by stimulating gut growth, enhancing nutrient absorption, and contributing to early immune function.

Hormones. Several hormones are present in colostrum (Figure 5): prolactin is present 280 μg/L in colostrum versus 15 μg/L in mature milk, glucagon occurs at 0.16 μg/L versus 0.01 in mature milk, and insulin is present at 65 μg/L versus 1 μg/L in mature milk (Blum and Hammon, 2000).

Insulin and glucagon regulate metabolic adaptation after birth, with insulin promoting growth and energy storage and glucagon maintaining blood glucose through glycogen breakdown. Higher concentrations of these components in colostrum compared with mature milk may reflect the calf's immediate need for metabolic support and rapid developmental transitions during the early postnatal period. However, previous work measuring plasma insulin dynamics after colostrum ingestion suggests that observed insulin may primarily result from endogenous secretion, as indicated by tissue mRNA expression (Hammon et al., 2000). In contrast, glucagon is most likely derived from colostrum ingestion and reflects the active gluconeogenesis occurring early in life (Hammon and Blum, 1998). The direct influence of prolactin is not fully understood; however, the combined effects of hormone presence may contribute to the improvement of substrates to enhance metabolism and regulatory functions (Hammon and Blum, 1998). Recently, researchers demonstrated that supplementing insulin to pharmacological levels in colostrum did not affect IgG absorption (Hare et al., 2023). Cortisol levels in colostrum are known to vary due to maternal stress around parturition, leading to its transfer into colostrum. Although not quantified in this review, cortisol may contribute to GI development in neonates by promoting digestive enzyme secretion; however, its specific effect is difficult to isolate given the influence of other colostral components (Blum, 2006).

Amino Acids and Proteins. Colostrum contain a greater proportion of protein-bound EAA and NEAA to mature milk (Figure 6; Zanker et al., 2000). Addition-

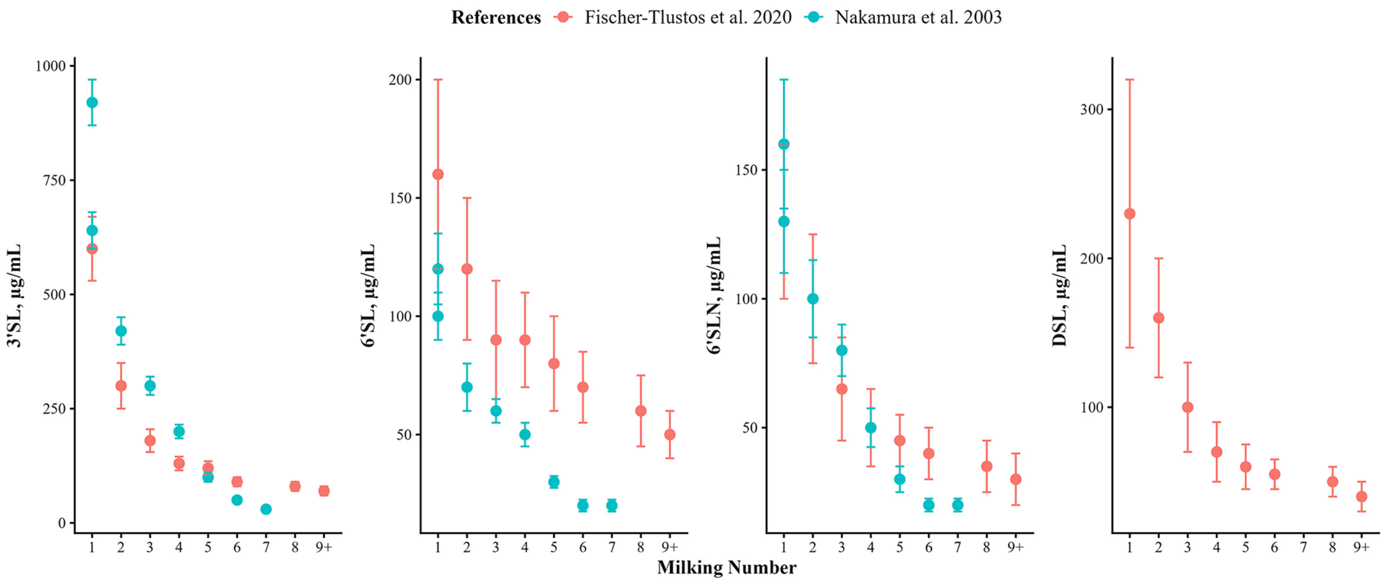

Figure 8. Oligosaccharide content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 2 studies included; Fischer-Tlustos et al., 2020; Nakamura et al., 2003). Components from milk extracted and presented here (in µg/mL) include 3′-sialyllactose (3′SL), 6′-sialyllactose (6′SL), 6′-sialyllactosamine (6′SLN), and disialyllactose (DSL). Error bars represent SE.

ally, selected AA implicated for the development of the neonatal calf: free Val is higher in colostrum at 46.5 µmol/L compared with 11.6 µmol/L in mature milk, and free Met is present at 6 µmol/L in colostrum and decreases by milking 5 to 3.2 µmol/L. In contrast, free Glu increases from 10 µmol/L in colostrum to 108.6 µmol/L in mature milk. Free His and Lys peak during transition milk at milking 3 (357.9 µmol/L) and milking 2 (4.5 µmol/L), respectively, before declining to 130.7 µmol/L and 3.3 µmol/L in mature milk. Additionally, free Tau is highest in colostrum at 193.7 µmol/L and decreases to 15.6 µmol/L in mature milk. Protein-bound AA are consistently greater in colostrum (milking 1), including His (16,424 vs. 4,665 µmol/L), Lys (57,815 vs. 15,486 µmol/L), Met (13,486 vs. 3,608 µmol/L), Thr (53,663 vs. 8,434 µmol/L), and Val (64,680 vs. 13,362 µmol/L), when compared with mature milk. Colostrum has 138 unique proteins and shares 514 proteins with mature milk (Nissen et al., 2017). Additionally, one protein of interest is lactoferrin, which ranges from 0.72 to 1.28 mg/mL in colostrum to 0.18 to 0.19 mg/mL in mature milk (Kehoe et al., 2007; Tortadès et al., 2023a).

Amino acids are important in early life for development because calves may not have them in abundance at birth. The unique proteins found in colostrum have many functions and are generally regarded to help establish the calf's immune system and aid in growth and development (Zanker et al., 2000; Nissen et al., 2017). Lactoferrin is considered an antimicrobial peptide and may contribute to maintaining a healthy gut in neonatal calves (Pakkanen and Aalto, 1997). Although present in different quantities depending on parity (greater in multiparous), they are still more concentrated in colostrum compared with mature milk (Tortadès et al., 2023a).

Cytokines. Levels of cytokines in colostrum are higher compared with mature milk (Figure 7): TNF-α is present at 5.00 to 956.55 µg/mL in colostrum versus 2.00 to 4.58 µg/mL in mature milk, IL-1β at 844.24 µg/mL versus 0.49 to 3.43 µg/mL, IL-6 at 77.08 µg/mL versus 0.06 to 0.22 µg/mL, and INF-γ at 261.37 µg/mL versus 0.00 to 0.21 µg/mL (Blum and Hammon, 2000; Hagiwara et al., 2000).

Fetal-type enterocytes are responsible for large molecule uptake in the first 24–48 h of life, including cytokine uptake (Guilloteau et al., 2009). At birth, the immune system is naive and possesses little to no circulating immune proinflammatory cytokines (Yamanaka et al., 2003a). After birth, an increase of IL-6, IL-1β, and TNF-α may have an impact on the immune system during the first week of life as seen circulating in calf serum after colostrum ingestion (Peetsalu et al., 2022). These cytokines work together to mediate inflammation in the event of an immune challenge. IL-6 helps stimulate the production of acute-phase proteins by the liver and supports B cell differentiation, contributing to the calf's ability to mount an initial immune defense (Tanaka et al., 2014). IL-1β acts as a potent mediator of inflammation, promoting fever and activating immune cells such as macrophages and neutrophils to respond to pathogens (Rider et al., 2011; Koh et al., 2018; Aarreberg et al., 2019). TNF-α supports the recruitment of immune cells to infection sites and enhances the permeability of vascular endothelium, enabling immune cell trafficking (Playford et al., 2000).

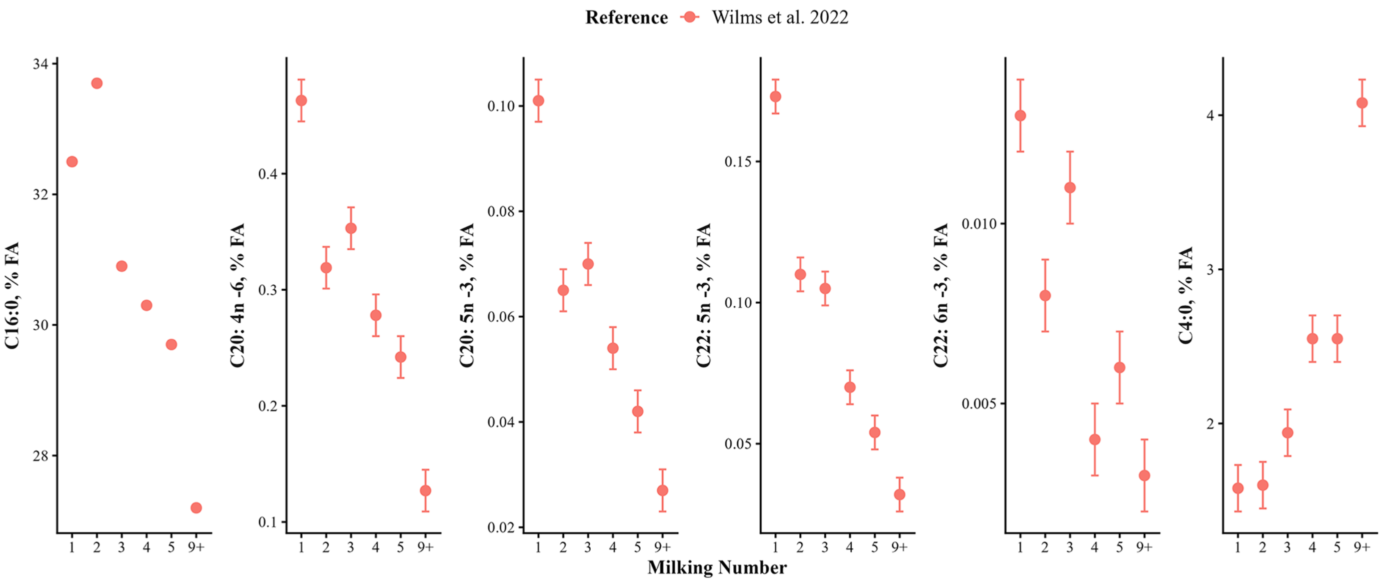

Figure 9. Selected FA content of bovine colostrum, transition milk, and mature milk over time extracted from a literature search from 2000 to 2025 (n = 1 study included; Wilms et al., 2022). Components from milk extracted and presented here (as % of total FA) include C4:0, C16:0, C20: 4n-6, C20:5n-3, C22:5n-3, and C22:6n-3. Error bars represent SE.

Alternatively, a high presence of inflammatory cytokine mRNAs may be deleterious to the neonate, causing impaired performance (Hagiwara et al., 2001). They stimulate immune function by enhancing mitogenic response of mononuclear immune cells (Yamanaka et al., 2003b).

Oligosaccharides. The most abundant oligosaccharides, sialyloligosaccharides, are found at significantly higher levels in colostrum compared with mature milk (Figure 8). Specifically, 3′-sialyllactose (3′SL) is present at 600 to 920 μg/g in colostrum versus 70 μg/g in mature milk, 6′-sialyllactose (6′SL) at 100 to 160 μg/g versus 50 μg/g, 6′-sialyllactosamine (6′SLN) at 130 to 160 μg/g versus 30 μg/g, and disialyllactose (DSL) at 230 μg/g versus 40 μg/g (Nakamura et al., 2003; Fischer-Tlustos et al., 2020).

Oligosaccharides remain present in the ileum and colon of the calf GIT (Fischer et al., 2018). In the same study identifying the location of oligosaccharide activity, heat treatment of natural bovine colostrum was found to enhance the presence of oligosaccharides reviewed here, although it did not significantly alter the overall ratio of specific oligosaccharides associated with beneficial effects in calves. Specifically, oligosaccharides in colostrum support *Bifidobacterium* by binding to pathogens most commonly affecting neonatal calves, including enterotoxigenic *Escherichia coli* (Martín et al., 2003). Other forms of oligosaccharides include mannan-, galacto- and fructo-oligosaccharides and have been investigated for their role in development and morbidity. In one study, mannan-oligosaccharide reduced IgG absorption when fed with colostrum (Brady et al., 2015). Other studies investigated forms of inactive oligosaccharides in nursing calves during the milk phase of life (after colostrum intake, assuming intake of natural sialyloligosaccharides) and saw improvements in reducing morbidity and increasing development (Gao et al., 2024; Gilbert et al., 2024). Overall, oligosaccharides in colostrum play a crucial protective role in the neonatal gut by promoting beneficial bacteria and preventing pathogen attachment, supporting early immune defense and gut health.

Fatty Acids. Fatty acid (**FA**) profiles change between colostrum and mature milk (Wilms et al., 2022). Selected FA are presented in Figure 9: palmitic acid (C16:0; PA) is higher at 32.5% in colostrum versus 27.2% in mature milk, and the same decreasing pattern is observed for arachidonic acid (C20:4n-6; ARA, 0.462% vs. 0.127%), eicosapentaenoic acid (C20:5n-3; EPA, 0.101% vs. 0.027%), docosapentaenoic acid (C22:5n-3; 0.173% vs. 0.032%), and docosahexaenoic acid (C22:6n-3; **DHA**, 0.013% vs. 0.003%), whereas butyric acid (C4:0; BA) shows the opposite trend, increasing from 1.58% in colostrum to 4.08% in mature milk. Overall, SFA constitute 69.9% of total FA in colostrum compared with 66.9% of total FA in mature milk. De novo FA accounts for 26.1% of total FA in colostrum versus 24.5% in mature milk, and total branched-chain FA make up 1.33% of total FA in colostrum compared with 1.52% in mature milk. Total *cis*-MUFA are 24.2% of total FA in colostrum versus 24.1% in mature milk, total *trans*-MUFA are 4.38% versus 3.87%, and total MUFA are 28.7% versus 27.9% in colostrum and mature milk, respectively. Additionally, total n-3 PUFA are 0.444% in colostrum compared with

0.305% in mature milk, and total PUFA are 3.22% in colostrum versus 2.62% in mature milk.

Fatty acids have a range of effects on neonatal calves ranging from gut maturation to anti-inflammatory properties, immune system development, gluconeogenesis and glucose homeostasis, and more (Hammon et al., 2012; Wilms et al., 2022). Docosahexaenoic acid (C22:6n-3) promotes brain maturation and immune system development during the early postnatal period. Docosapentaenoic acid (C22:5n-3; DPA) contributes to nervous system development and may function as a storage form of DHA and eicosapentaenoic acid (**EPA**; C20:5n-3), both of which play roles in regulating inflammation, which is especially important after birth. Arachidonic acid (C20:4n-6) and EPA are involved in the synthesis of eicosanoids, thereby supporting immune maturation in neonatal calves. Palmitic acid (C16:0) serves as a key energy source and a precursor for brain lipids essential for neural and immune development. Additionally, butyric acid (C4:0) is speculated to play a role in maintaining intestinal permeability during the first days of life, thereby facilitating the absorption of Ig through pinocytosis. However, one study investigated the effects of butyrate added to a colostrum replacer and observed reduced passive transfer of immunity (Hiltz and Laarman, 2019). Overall, trends of FA presence in colostrum suggest importance for immune development and function.

Vitamins and Minerals. In colostrum, reported vitamin concentrations include approximately 4.9 μg/g of retinol, 2.92 μg/g of tocopherol, 77.17 μg/g of fat for vitamin E, 0.9 μg/g of thiamin, 4.55 μg/g of riboflavin, and 0.6 μg/g of vitamin B12 (Kehoe et al., 2007). For minerals, calcium ranges from 54.20 to 55.71 mmol/L in colostrum versus 28.30 to 33.50 mmol/L in mature milk, phosphorus ranges from 41.91 to 52.80 mmol/L versus 23.00 to 30.00 mmol/L, magnesium ranges from 12.10 to 13.44 mmol/L versus 4.40 to 5.60 mmol/L, sodium is at 37.83 mmol/L in colostrum, potassium is 36.31 mmol/L, zinc is at 272.13 μmol/L, iron is at 12.51 μmol/L, copper is at 3.34 μmol/L, and manganese is at 0.1 μmol/L (Kehoe et al., 2007; Tsioulpas et al., 2007; Valldecabres and Silva-del-Río, 2022).

Calves are born vitamin deficient due to lack of passage across the placenta and rely on ingestion via colostrum (NASEM, 2021). Not having a vitamin supply may compromise plasma status of the neonatal calf (Blum et al., 1997). Vitamins are critical for establishing many important systems in a calf for the first week of life (Blum et al., 1997). The effects of vitamin A and E are important for leukocyte function (Eicher et al., 1994). All vitamins are critical for health and growth of the neonatal calf and for providing substrates for important metabolic functions to take place (Godden, 2008). Minerals are important for the development of structures including bone, as well as overall development and growth (NRC, 2001). In older animals, minerals play an essential role in structures, enzyme cofactors and activators, electrolyte and fluid balance, and hormone synthesis and immune function (Spears, 2000; Weiss and Spears, 2006). Recently, researchers investigated the effects of vitamin and mineral supplementation in addition to feeding colostrum replacer on copper, selenium, and vitamin D status in calves (Hurlbert et al., 2024). The supraphysiological supply of vitamins and minerals may improve growth and development of the calf, but additional studies are needed to address differences in the existing supply of minerals and the lack of comprehensive knowledge regarding mineral and trace mineral status of neonatal calves.

Enzymes and Microbiome. Although various enzymes are present in colostrum, lack of data on their relative abundance narrows this review to a brief overview of their general roles. Lactoperoxidase and lysozyme are well documented for supporting gut health (Pakkanen and Aalto, 1997). Lysozyme plays a protective role against pathogens and has been positively correlated with serum concentrations in calves (Paulík et al., 1985). Trypsin inhibitors present in colostrum help shield immune components from proteolytic degradation (Laskowski Jr. and Laskowski, 1951). These inhibitors have been associated with IgG concentrations in Jersey cows and influence transfer of passive immunity (Quigley et al., 1995a). Supplementing colostrum with trypsin inhibitors from soy has also been shown to increase serum IgG (Quigley et al., 1995b). Although calves naturally have low trypsin activity at birth, early colostrum feeding within 24 h minimizes the need for inhibitors. However, the presence of trypsin inhibitors still plays a key role in preserving IgG structure and limiting proteolysis in the neonatal gut. More recently, research has shown that enzymatic and bioactive activity in colostrum is higher when it is not heat-treated (Mann et al., 2020).

CONCLUSIONS

Bovine colostrum plays a critical role in the early development of neonatal calves by providing immunoglobulins, growth factors, hormones, AA, and other bioactive compounds that support immune function, growth, and gastrointestinal development. Compared with the extensive body of work on IgG, the literature describing secondary colostrum components is relatively limited, and their dynamics across successive milkings remain poorly characterized. This gap highlights an important opportunity for further research to better understand the dynamics and physiological roles of these components. Continued investigation could enhance our ability to contextualize their implications for calf development and inform more targeted management strategies, such as extended colostrum feeding and the use of transition milk in neonatal calves, ultimately improving colostrum management and neonatal outcomes.

NOTES

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Nonstandard abbreviations used: 3′SL = 3′-sialyllactose; 6′SL = 6′-sialyllactose; 6′SLN = 6′-sialyllactosamine; DHA = docosahexaenoic acid; DSL = disialyllactose; EFG = epidermal growth factors; EPA = eicosapentaenoic acid; F = free; FA = fatty acids; GH = growth hormone; GIT = gastrointestinal tract; PB = protein bound; TGF = transforming growth factors.

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